# Prices First, Portfolios Later: The Transmission of Federal Reserve Communication to Asset Prices and Household Portfolios

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Working paper

## 1 2. Theoretical Framework

Our framework rests on three pillars. Each pillar maps to a measurement discipline, and together they generate the testable implications that structure the paper. We state the theory first because the empirical design — including two deliberate nulls — is derived, not discovered.

### 1.1 2.1 Pillar 1 — Rational expectations and the market's inversion rule

Under rational expectations, the announcement-day price response to an FOMC statement is the market's Bayesian posterior update about the policy path and the state of the economy, given everything the market knew before the statement. This has a consequence that drives the whole paper: the *market's* reaction is the sufficient statistic for the news content of the communication. Any analyst-side measure of what the statement "said" — dictionary scores, LLM scores, human readings — is an attempt to reconstruct that news from the text alone, and is informative about the text, not about the market's information set.

The central bank's statement enters this inversion twice. First, through the *decision* channel: the target-rate innovation relative to the market's prior. Second, through the *information* channel: the statement reveals the committee's private assessment of the economy, which is valuable to investors independent of the current decision (Coibion, Gorodnichenko, and Ulig 2022; Jarociński and Karadi 2020). We call the market's extraction of this second component the **central-bank-information (CBI) shock**. CBI is, by construction, the market's inference about the central bank's information set — not a property of the text. This distinction disciplines every measurement choice that follows: a shock variable that correlates with the *text* is not the market's inversion outcome, and the empirical content of Section 7 rests precisely on keeping text-measures and inversion-measures separate.

### 1.2 2.2 Pillar 2 — From measurement window to shock decomposition

Kuttner (2001) established the measurement principle: identify policy news from asset-price movements in a narrow window around the announcement, because those prices aggregate the market's information faster than any textual analysis. Bauer and Swanson (2023) sharpened it into an oracle argument — under RE, the

*true* policy surprise equals the market's price response up to a risk-premium correction, so any independently measured "surprise" that disagrees with prices is mismeasured, not the market.

Jarociński and Karadi (2020) then showed that the price response itself carries a sign structure that separates two economically distinct events: a conventional monetary shock (MP) and an information shock (CBI) in the Coibion–Gorodnichenko–Ulig sense — the central bank revealing information about the economy rather than about its reaction function. MP and CBI move rates in the same direction but equity prices in opposite directions.

We implement the JK idea with a deliberately transparent inversion rule. Let $\Delta S$ be the same-day CRSP value-weighted return and $\Delta i$ the policy-rate innovation embedded in the surprise. We assign

$$\mathrm{MP}_m = \Delta i_m \cdot \mathbb{1}[\mathrm{sign}(\Delta i_m) \neq \mathrm{sign}(\Delta S_m)], \qquad \mathrm{CBI}_m = \Delta i_m \cdot \mathbb{1}[\mathrm{sign}(\Delta i_m) = \mathrm{sign}(\Delta S_m)].$$

The sign restriction *is* the inversion rule: when the equity market moves against the rate move, the market is pricing a conventional shock; when they move together, the dominant news is information. This construction is coarse relative to full JK high-frequency identification, and Section 8 shows the flow-chapter results are robust to the refinement — but its transparency is a feature: every observation's classification is verifiable by inspection, which matters for a paper whose second half is about measurement discipline.

### 1.3 2.3 Pillar 3 — Arbitrage pricing and the risk-compensation channel

If communication shocks are risks, they must be compensated. The arbitrage-pricing view (Ross 1976) predicts that the cross-section of *portfolio* risk premia loads on the shock factors: a factor priced for investors must appear in expected returns, and — in the mutual-fund margin we observe — in reallocation flows as investors traverse the risk ladder (Metcalfe, Bhojraj, and Mozes 2020). Pillar 3 delivers the paper's portfolio predictions: a factor whose realization is observable at the announcement must show up in (i) risk-premium pricing and (ii) fund flows that re-tilt portfolios along the risk ladder.

### 1.4 2.4 Testable implications

| # | Implication | Pillar | Where tested |
|---|---|---|---|
| T1 | MP and CBI both move announcement-day prices, with opposite equity signs | P1+P2 | §4 |

| # | Implication | Pillar | Where tested |
|---|---|---|---|
| T2 | Statement *text* (tone) carries independent price information beyond shocks, concentrated where forward guidance binds | P1+P2 | §4 |
| T3 | Neither shock moves monthly fund flows once the price channel has cleared | P3 | §5 |
| T4 | Flow responses follow the risk ladder: safe-asset inflows on hawkish information, risky-asset outflows | P3 | §6 |
| T5 | When the communication paradigm shifts, dictionary-based text measures fail in a characteristic way (resolution loss, not bias) | P1 | §7 |

Two implications are deliberately *null* — T3, and the flow-side half of T2. Following the contribution discipline fixed with the advising team: a theory that predicts a null where the heuristic expects an effect is more falsifiable than one that predicts an effect everywhere, and the nulls are load-bearing for the paper's mechanism (prices clear the information first; portfolios adjust later, along the risk ladder, to the *risk* content rather than the *words*). # 4. Prices: What the Statement Text Adds to the Shocks

This section tests T1 and T2. The dependent variables are same-day changes for 10-year and 2-year Treasury futures and same-day equity returns (CRSP value-weighted and S&P 500), observed for the 161 meetings from 2006 to 2026 (150 with CRSP coverage). The text measure is dictionary tone per 100 words (LM hawkish-minus-dovish, `tone100`). Regime indicators mark the forward-guidance era (December 2008 through December 2015, 56 meetings), the 2020–2022 zero-rate window, and the post-2022 tightening cycle. All regressions are OLS with heteroskedasticity-robust standard errors.

## 1.5 4.1 The regime structure: tone matters only under forward guidance

Table 4.1 reports the tone coefficient by asset and regime (Panel A: unconditional; Panel B: the FG regime split).

**Cross-model audit note.** Every statement score used in this section passed the five-layer hallucination audit (Appendix A): the LLM-vs-dictionary divergences that define Section 7 were confirmed across four vendor families (sign agreement 100% on the 11 divergence statements), and one one-off scoring error (2011-09-21) was caught and corrected by the cross-model committee.

*[TABLE 4.1 — tone coefficient by asset and regime, from `price_results_v1.json` Panel B: CRSP VW FG −1.119 (p=0.004, R²=0.141, n=56); S&P 500 FG −1.027 (p=0.006, R²=0.130); CRSP VW non-FG +0.037 (p=0.803); TY10 FG +0.044 (p=0.134); TY10 non-FG −0.007 (p=0.317); TY2 all regimes p>0.32]*

Three facts stand out. First, in the FG era a one-standard-deviation hawkish shift in tone (about 12 points) is associated with a −1.12 percentage-point same-day equity return — economically large, and restricted to the FG regime: outside it, the tone coefficient is indistinguishable from zero (+0.04, p=0.80). Second, the sign pattern across assets is exactly what a hawkish text signal implies: equity down, and — in the interaction specification — Treasury yields up (tone×FG on TY10: +0.050, p=0.030). The same words push discount rates up and cash flows down. Third, the short end (TY2) shows no tone response at any horizon: tone acts through the *medium-horizon path* expectations that forward guidance manages, not through the near-term policy rate.

## 1.6 4.2 Horse race: tone survives the shocks

Is tone just a noisy proxy for the shocks? No. Table 4.2 (Panel C) puts tone, MP, and CBI in the same regression for the FG subsample. On CRSP VW:

$$\Delta S_m = -1.21\,\text{tone}_m - 1.25\,\text{MP}_m + 1.93\,\text{CBI}_m, \quad R^2 = 0.48,\ n = 56,$$

with all three coefficients significant (tone p<0.001, MP p<0.001, CBI p=0.004). The sign structure is the theory's: conventional shocks (MP) depress equity; information shocks (CBI) *raise* it — the market reads hawkish information as evidence of a stronger economy (Jarociński and Karadi 2020) — and tone carries independent, incremental information about the same-day return beyond both shocks. That a per-100-words dictionary score, orthogonal to the surprise vector by construction, adds 48-percent-of-variance explanatory power jointly with the shocks is the strongest form of T2: the statement's *text* is a separate information channel, not a reflection of the decision.

*[TABLE 4.2 — FG horse race, all four assets; note CBI's positive equity loading and MP's negative one, both consistent with JK; tone loads negatively on equity and positively on TY10 in the interaction spec]*

## 1.7 4.3 Orthogonalization as a no-op, and why that answers the question

The advising team asked that tone be orthogonalized to Target and Path before entering the flow regressions of Section 5. It was: orthogonalized tone reproduces raw tone to the fourth decimal (flow-side p=0.635 raw vs 0.625 orth; incremental R² of the orthogonalization 0.4%), because the dictionary score is constructed

from the statement text after the decision window and shares almost no variance with the surprise vector. We report the orth spec alongside the raw one throughout — and we flag the deeper point: the near-orthogonality is not a nuisance, it is what makes tone a *separate* measurement channel. A text measure that were highly correlated with the market-priced surprises would be redundant with them; the dictionary's near-independence is precisely why its FG-regime price effect is informative.

### 1.8 4.4 Where this leaves the price side

The announcement-day evidence supports T1 and T2: both shock components move prices with the JK sign structure, and the text carries incremental price information concentrated exactly in the regime where the committee's words were the policy instrument (forward guidance). What the text does *not* do — per the framework's deliberate null — is move monthly flows. That is Section 5. # 5. Flows: The Ruling Null

T3 predicts that neither shock component moves monthly fund flows: prices have already cleared the information by the time a monthly flow window opens. This section tests that prediction — and reports the paper's first deliberate null.

### 1.9 5.1 Design

The flow panel covers 1,099 fund-month observations across seven Morningstar asset classes (government bonds, corporate bonds, real assets, large-cap / developed / emerging equity, small-cap equity; 157 meeting months, point-in-time classification, server-side aggregation, flows capped at 50% of TNA, windows sharing no calendar months with the gap rule `<=3` months). The specification follows the flow chapter: class-level net flow rate on the shock pair, fund size, and flow volatility, with fund fixed effects absorbed at the class-month aggregation level.

### 1.10 5.2 The null holds everywhere

Adding statement tone to the flow regressions changes nothing (models_v2, October 20 run): tone enters at p = 0.64 raw and p = 0.63 orthogonalized in the same-month window; in the forward-guidance subsample, p = 0.94. The LLM-based score behaves identically (p = 0.17). No statement-text measure — dictionary or model — has flow explanatory power at any horizon we test, while the same measures move announcement-day prices by economically large margins (Section 4). The null is not an artifact of one measure, one window, or one asset class.

We emphasize what this null is *not*. It is not evidence that households ignore the Fed: Section 6 shows the same funds reallocate sharply along the risk ladder in response to the information content of the same announcements. It is evidence about *timing and content*: by the time monthly flows are measured, the words have already been priced, and what remains for flows to respond to is the risk content, not the words.

### 1.11 5.3 The shock components' direct flow effects

The shocks themselves are likewise silent in the same-month window: MP enters at p = 0.92 and CBI at p = 0.29 (government bonds; the other six classes are in Appendix Table A3). The only significant flow-level

interaction involving MP arrives in the zero-rate regime (Section 6, Table 6.2) — and it is a *risk-taking* effect, not an information effect, consistent with the theory's separation of channels. # 6. The Risk-Taking Channel: De-Risking on Hawkish Information

T4 predicts flow reallocation along the Metcalfe-style risk ladder in response to the information component of policy news. The finding reverses the paper's original H5 hypothesis — announced in the proposal as risk-*taking* on expansionary surprises — and the reversal is itself informative: what households and institutions respond to is the *direction of new information about the economy*, not the sign of the policy move.

### 1.12 6.1 Risk ladder interaction

With each fund class mapped to a 1–7 ordinal risk rank (government bonds = 1 … small-cap equity = 7), the interaction of the risk rank with the CBI shock is negative and significant: **Risk × CBI = −0.066 (p = 0.036, n = 1,099)**, while Risk × MP is a precise zero (+0.005, p = 0.88). The CBI main effect is positive (+0.332, p = 0.019) — the low-risk anchor class *receives* flows on information shocks — so the interaction says: the higher the class sits on the risk ladder, the more it loses on the same information event. One unit of hawkish information news shifts roughly 0.07 pp of monthly flow from the riskiest class toward the safest, per unit of shock. This is **de-risking on hawkish information**, exactly the ordering T4 predicts.

The Wald test cannot reject equality of the two interaction coefficients (p = 0.11); we therefore claim the asymmetry in *sign* (MP flows are risk-neutral; CBI flows are risk-ordered), not in magnitude.

### 1.13 6.2 Sign asymmetry: positive news does the work

Splitting CBI by sign sharpens the channel (models_v2): **Risk × CBI(+) = −0.088 (p = 0.020)**; Risk × CBI(−) = −0.055 (p = 0.10). Hawkish information — the market telling the investor that the Fed knows something bad — drives the ladder. Dovish information produces a weak, statistically indistinguishable version of the same slope. The economics: funds do not chase the *policy* on good news; they flee *risk* on bad news. The asymmetry is the portfolio-side footprint of the asymmetry that Badarinath–Frydman-style and homestead asset-pricing work has long documented for prices, here appearing one step later, in quantities.

### 1.14 6.3 The zero-rate exception

In the 2020–2022 zero-rate window, the *conventional* component acquires a flow effect: **MP × ZLB = −1.505 (p = 0.005, n = 1,099, $R^2$ = 0.050)** — a one-unit MP shock at the ZLB pushes roughly 1.5 pp of monthly flow out of the average class, concentrated in the safe classes that normally absorb the CBI response (CBI × ZLB is a precise zero, p = 0.78). The reading follows the exposition in Section 2: at the ZLB a conventional surprise is an *exit option* event — the committee has no room to offset a hawkish surprise with further cuts, so even small conventional news is portfolio-relevant, and it operates through the rescue-capacity channel rather than the information channel (whose interaction is null precisely when the level channel binds).

## 1.15 6.4 Putting Sections 4–6 together

The three sections trace one mechanism in three acts. *Prices* clear the statement's information on the announcement day — shocks with JK signs, tone with incremental FG-regime information (Section 4). *Monthly flows* ignore the words entirely — the ruling null (Section 5). *Flows* respond only to the risk content of the information, along the ladder, asymmetrically, with a ZLB exception that reveals the conventional channel working through rescue capacity rather than news (Section 6). Prices first; portfolios later; and what reaches portfolios is not the sentence — it is the risk.